

## CELL BIOLOGY

# Chemoattractant Signaling in *Dictyostelium*: Adaptation and Amplification

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**The social amoebae *Dictyostelium discoideum* has long proved a powerful model organism for studying how cells sense and interpret chemoattractant gradients. Because of the rich behavior observed in its response to chemoattractants, as well as the complex nature of the signaling pathways involved, this research has attracted and benefited from the use of theoretical models. Recent quantitative experiments provide support for a popular model: the local excitation, global inhibition mechanism of gradient sensing. Here, I discuss these findings and suggest some important open problems.**

## Introduction

Chemotaxis, the ability to sense chemical gradients and use this information to direct motion, is crucial for the proper function of nearly all organisms. Chemotactic cells show a remarkable ability to interpret shallow gradients, whereby differences in receptor occupancy as small as 1% lead to polarization and subsequent motion up a gradient. This precision is attributed to their ability to adapt—to adjust their sensitivity in response to spatially uniform changes in chemoattractant concentration—and to amplify the small gradient of receptor occupancy. Two papers (1, 2) combine quantitative experiments using the model chemotactic amoebae, *Dictyostelium discoideum*, in microfluidic chambers, and computer simulations to shed light on these two processes. In doing so, they help to discern among competing signaling topologies.

## Adaptation

How cells adapt to persistent changes in chemoattractant has been a long-standing question in biology that has attracted much interest from theoretical biologists. Two general models have been proposed, and subsequent analysis has shown that these models represent the only two network topologies that achieve adaptation (3). One class of models assumes that adaptation comes about through a negative feedback loop (4). If the feedback is based on the integral of the difference between desired and achieved

response—a type of feedback known by engineers as integral control—then perfect adaptation is achieved (5). This property is robust, meaning that it does not depend on precise values of parameters, as demonstrated experimentally in bacteria (6). The second class of models posits that adaptation results from the action of complementary excitation and inhibition processes that together regulate a response regulator. Because the scheme involves direct connections from receptor signaling to the two complementary processes, it is also known as an incoherent feedforward loop. Although not well appreciated, the incoherent feedforward loop topology can be recast as an integral controller feedback loop (7, 8) and has its associated robustness properties (9).

Originally proposed to explain the adaptive response to chemoattractants of *Escherichia coli* cells (10), which sense gradients temporally, the model has been modified to explain the static spatial sensing property of *Dictyostelium* and other eukaryotic cells, by assigning differing spatial characteristics to these two processes. The resultant local excitation, global inhibition paradigm (LEGI) is now widely accepted (9, 11–13). However, to date, experimental evidence for the LEGI mechanism has been mostly circumstantial. Measurements of the gradient response of *Dictyostelium* cells support one of the LEGI model predictions: The spatial distribution in the steady-state response depends on the ratio between the local and global receptor signals (14). Observations of the spatial distribution of intracellular signals after addition and removal of chemoattractant also corroborate the presence of competing processes that are locally regulated by the receptor signal and show that

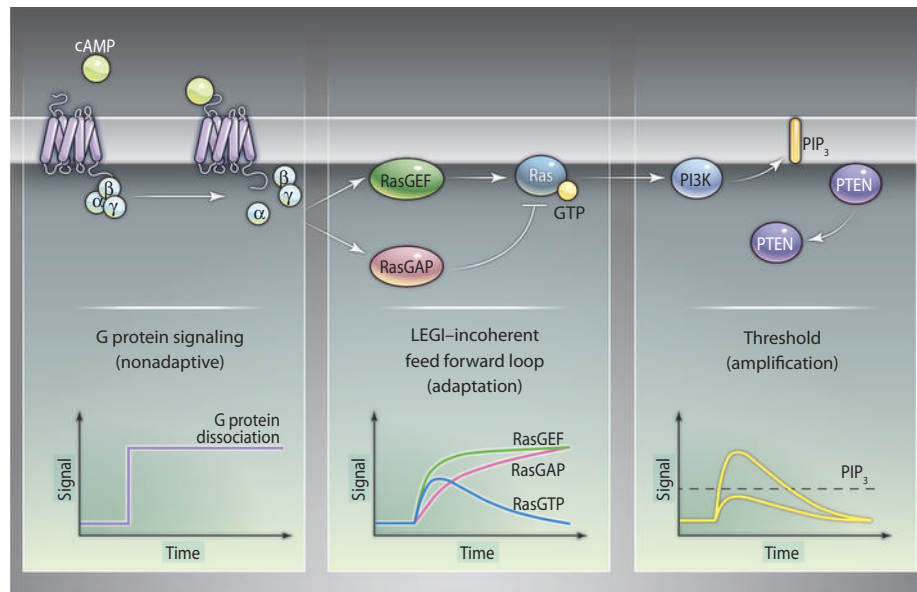
the excitation process occurs at a slower rate than the slow inhibition process (14, 15).

By measuring the receptor-mediated activation of the guanosine triphosphatase (GTPase) RasG, Takeda and colleagues have now provided further evidence for the presence of a LEGI mechanism (1). In *Dictyostelium*, one of the earliest downstream signals after chemoattractant binding to heterotrimeric guanine nucleotide-binding protein (G protein)-coupled receptors is the activation of Ras proteins (Fig. 1) (16). This activity can be observed by tracking the fluorescently tagged Ras-binding domain of human Raf1, which binds to RasG, one of the *Dictyostelium* Ras proteins. Ras proteins cycle between active guanosine triphosphate (GTP)-bound and inactive guanosine diphosphate (GDP)-bound states (16–18). Activation is regulated by guanine nucleotide exchange factors (GEFs), and inactivation is regulated by GTPase-activating proteins (GAPs) that stimulate the hydrolysis of the bound GTP to GDP. Using microfluidic devices to stimulate cells, Takeda *et al.* demonstrated that RasG activation dynamics closely follow the LEGI-incoherent feedforward loop model (Fig. 1), under the assumption that the chemoattractant receptor signal stimulates the activity of the RasG-GEF (the exciter) and RasG-GAP (the inhibitor). Moreover, they varied chemoattractant concentration and measured the corresponding temporal dose responses. A prediction of the LEGI mechanism is that the time at which the peak response occurs and the time to adapt will both be shorter as the concentration of chemoattractant is increased. This effect occurs because, in the LEGI mechanism, it is the concentration of the inhibitor that determines the time scale of adaptation. Because receptor occupancy regulates the inhibitor, higher concentrations of chemoattractant lead to higher inhibitor concentrations and faster peak and adaptation times. This prediction can be verified using an interactive applet of the LEGI mechanism, available at <http://stke.sciencemag.org/feature/legi/> (12). The observed RasG activation follows this prediction exactly.

Takeda *et al.* fit their data to competing models based on LEGI-incoherent feedforward loop and feedback control topologies and showed a better match between simulation and experiments using the LEGI-incoherent feedforward loop model. Although they did not consider the cellular response to chemoattractant gradients, their model

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**Fig. 1. Adaptation and amplification of the chemoattractant response in *Dictyostelium*.** The chemoattractant cyclic adenosine monophosphate (cAMP) binds to G protein–coupled receptors and causes the G protein  $\alpha$  and  $\beta\gamma$  subunits to dissociate in a persistent manner (bottom, left panel). One of the earliest downstream signals is RasGTP activation, mediated by RasGEF and RasGAP. RasGTP activity adapts perfectly (blue line; bottom, middle panel), in agreement with a LEGI model in which RasGEF acts as the excitation (green line; bottom, middle panel), and RasGAP as the inhibitor (red line; bottom, middle panel). Ras activates the PI3K signaling pathway, which can be monitored using  $\text{PIP}_3$ -specific biosensors. The observed behavior suggests the presence of a threshold (dotted line; bottom, right panel). Signals above the threshold elicit responses (dotted line; bottom, right panel); subthreshold signals do not (dashed line; bottom, right panel).

fits the spatial requirements of the LEGI model because the *Dictyostelium* RasGAP, NF1, is globally distributed, and the RasG-specific GEF, RasGEFR, is believed to be locally activated.

### Amplification

In *Dictyostelium*, activation of Ras proteins subsequently leads to the production of phosphatidylinositol 3,4,5-trisphosphate ( $\text{PIP}_3$ ), mediated by transient increases in phosphoinositide 3-kinase (PI3K) activity and a temporally coincident loss of phosphatase and tensin homolog deleted from chromosome 10 (PTEN) from the membrane (18). Wang and colleagues used the  $\text{PIP}_3$ -specific biosensor PHcrac-GFP (in complex with green fluorescent protein) to investigate the dynamic response of this downstream system when subjected to changes in chemoattractant concentration in a microfluidic device (2). As previously observed, stimulation with chemoattractant caused transient increases in  $\text{PIP}_3$ . However, by tracking the response of individual cells, Wang *et al.* revealed that this increase is seen only in a fraction of cells and that

this fraction increased as the concentration of chemoattractant also increased. Thus, two distinct populations exist at any given concentration of chemoattractant. These observations point to the presence of a threshold of activation, together with a stochastic perturbation that alters the relation between the threshold value and the basal or adapted value of the response regulator and thus causes only some cells to respond (Fig. 1).

Previously, models have been proposed in which the presence of a switchlike circuit downstream of the LEGI module acts to discriminate between sub- and superthreshold levels in the response regulator (9, 19, 20). This threshold serves as an amplification step that accentuates the spatial differences sensed by the LEGI mechanism and fixes one of its deficiencies: Alone, the LEGI mechanism provides no amplification. Wang *et al.* use such a model to fit their data and find agreement between simulations and experimental results.

Wang *et al.* also provide further evidence that adaptation is achieved through a LEGI–incoherent feedforward loop scheme, rather than integral feedback. Ramp-like increases

in chemoattractant provide a means to discriminate between the incoherent feedforward loop topology of the LEGI mechanism and the integral feedback network. As taught to all control engineering undergraduates, a system that uses an integral control feedback loop that is subjected to a ramp increase in stimulus amount will not adapt perfectly, but instead settle to a constant value at steady state. Experiments in *E. coli* have demonstrated this behavior and have given further evidence that bacteria employ integral control (21). In contrast, when Wang *et al.* stimulated *Dictyostelium* cells with a ramp increase in chemoattractant, cells adapted to this increasing concentration, as predicted by the LEGI model (22).

### Conclusion

Together, these two papers provide convincing support that *Dictyostelium* cells sense and interpret gradients using the combination of a LEGI and an amplification step. Because the amplification step was observed downstream ( $\text{PIP}_3$ ) of the RasG signal, it is tempting to suggest that RasG acts as the response regulator in the LEGI mechanism, and that the PI3K signaling circuit as the amplifier (Fig. 1). However, there are a number of open questions that need to be addressed before we can accept this model unequivocally. First, is the activity or abundance of RasGAP increased by changes in receptor occupancy, as required by the LEGI model? At present, there is no direct evidence for this. As acknowledged by Takeda *et al.*, it is possible that adaptation could occur upstream of RasG—for example, if RasGEF activity were itself adaptive. Fluorescence resonance energy transfer (FRET) studies of the G protein have shown that the receptor-mediated dissociation between G protein subunits is persistent and does not adapt (23), but there is no current method of testing whether these subunits continue signaling after dissociation (Fig. 1). Second, does RasG activation have the same degree of diversity at the single-cell level as was observed in  $\text{PIP}_3$ ? If true, it would argue that threshold-dependent amplification occurs upstream of PI3K signaling. One hint that this is not so is the response of cells treated by the actin inhibitor latrunculin. Previous measurements of the spatial response have shown that amplification is seen even in latrunculin-treated cells, but that the degree of amplification increases greatly with an intact cytoskeleton (14). Takeda *et al.* found no difference in the response of cells with

or without actin demonstrating that at least some of the amplification occurs downstream of RasG. Third, what is the source of the amplification? We and others proposed models in which a LEGI mechanism senses and interprets the gradient and uses this information to bias the triggering of a downstream excitable network (24–26). In addition to providing great amplification, this scheme accounts for the spontaneous activity of unstimulated cells. Moreover, it can explain some other aspects of the  $\text{PIP}_3$  response, such as the presence of secondary patches of high activity that occur after chemoattractant stimulus (27). Determining whether such a system is indeed active in cells will also require the combination of careful quantitative experiments coupled to mathematical models used in both these papers.

### References

1. K. Takeda, D. Shao, M. Adler, P. G. Charest, W. F. Loomis, H. Levine, A. Groisman, W.-J. Rappel, R. A. Firtel, Incoherent feedforward control governs adaptation of activated Ras in a eukaryotic chemotaxis pathway. *Sci. Signal.* **5**, ra2 (2012).
2. C. J. Wang, A. Bergmann, B. Lin, K. Kim, A. Levchenko, Diverse sensitivity thresholds in dynamic signaling responses by social amoebae. *Sci. Signal.* **5**, ra17 (2012).
3. W. Ma, A. Trusina, H. El-Samad, W. A. Lim, C. Tang, Defining network topologies that can achieve biochemical adaptation. *Cell* **138**, 760–773 (2009).
4. N. Barkai, S. Leibler, Robustness in simple biochemical networks. *Nature* **387**, 913–917 (1997).
5. T. M. Yi, Y. Huang, M. I. Simon, J. Doyle, Robust perfect adaptation in bacterial chemotaxis through integral feedback control. *Proc. Natl. Acad. Sci. U.S.A.* **97**, 4649–4653 (2000).
6. U. Alon, M. G. Surette, N. Barkai, S. Leibler, Robustness in bacterial chemotaxis. *Nature* **397**, 168–171 (1999).
7. J. Krishnan, P. A. Iglesias, Analysis of the signal transduction properties of a module of spatial sensing in eukaryotic chemotaxis. *Bull. Math. Biol.* **65**, 95–128 (2003).
8. O. Shoval, U. Alon, E. D. Sontag, Symmetry invariance for adapting biological systems. *SIAM J. Appl. Dyn. Syst.* **10**, 857–886 (2011).
9. A. Levchenko, P. A. Iglesias, Models of eukaryotic gradient sensing: Application to chemotaxis of amoebae and neutrophils. *Biophys. J.* **82**, 50–63 (2002).
10. D. E. Koshland Jr., A response regulator model in a simple sensory system. *Science* **196**, 1055–1063 (1977).
11. C. A. Parent, P. N. Devreotes, A cell's sense of direction. *Science* **284**, 765–770 (1999).
12. B. Kutscher, P. Devreotes, P. A. Iglesias, Local excitation, global inhibition mechanism for gradient sensing: An interactive applet. *Sci. STKE* **2004**, pl3 (2004).
13. A. Jilkine, L. Edelstein-Keshet, A comparison of mathematical models for polarization of single eukaryotic cells in response to guided cues. *PLOS Comput. Biol.* **7**, e1001121 (2011).
14. C. Janetopoulos, L. Ma, P. N. Devreotes, P. A. Iglesias, Chemoattractant-induced phosphatidylinositol 3,4,5-trisphosphate accumulation is spatially amplified and adapts, independent of the actin cytoskeleton. *Proc. Natl. Acad. Sci. U.S.A.* **101**, 8951–8956 (2004).
15. X. Xu, M. Meier-Schellersheim, J. Yan, T. Jin, Locally controlled inhibitory mechanisms are involved in eukaryotic GPCR-mediated chemosensing. *J. Cell Biol.* **178**, 141–153 (2007).
16. H. Kae, A. Kortholt, H. Rehmann, R. H. Insall, P. J. Van Haastert, G. B. Spiegelman, G. Weeks, Cyclic AMP signalling in *Dictyostelium*: G-proteins activate separate Ras pathways using specific RasGEFs. *EMBO Rep.* **8**, 477–482 (2007).
17. P. G. Charest, R. A. Firtel, Feedback signaling controls leading-edge formation during chemotaxis. *Curr. Opin. Genet. Dev.* **16**, 339–347 (2006).
18. K. F. Swaney, C. H. Huang, P. N. Devreotes, Eukaryotic chemotaxis: A network of signaling pathways controls motility, directional sensing, and polarity. *Annu. Rev. Biophys.* **39**, 265–289 (2010).
19. P. A. Iglesias, A. Levchenko, Modeling the cell's guidance system. *Sci. STKE* **2002**, re12 (2002).
20. P. A. Iglesias, P. N. Devreotes, Navigating through models of chemotaxis. *Curr. Opin. Cell Biol.* **20**, 35–40 (2008).
21. T. S. Shimizu, Y. Tu, H. C. Berg, A modular gradient-sensing network for chemotaxis in *Escherichia coli* revealed by responses to time-varying stimuli. *Mol. Syst. Biol.* **6**, 382 (2010).
22. J. Krishnan, Effects of saturation and enzyme limitation in feedforward adaptive signal transduction. *IET Syst. Biol.* **5**, 208–219 (2011).
23. C. Janetopoulos, T. Jin, P. N. Devreotes, Receptor-mediated activation of heterotrimeric G-proteins in living cells. *Science* **291**, 2408–2411 (2001).
24. I. Hecht, D. A. Kessler, H. Levine, Transient localized patterns in noise-driven reaction-diffusion systems. *Phys. Rev. Lett.* **104**, 158301 (2010).
25. Y. Xiong, C. H. Huang, P. A. Iglesias, P. N. Devreotes, Cells navigate with a local-excitation, global-inhibition-biased excitable network. *Proc. Natl. Acad. Sci. U.S.A.* **107**, 17079–17086 (2010).
26. P. A. Iglesias, P. N. Devreotes, Biased excitable networks: How cells direct motion in response to gradients. *Curr. Opin. Cell Biol.* **24**, 10.1016/j.ceb.2011.11.009 (2012).
27. M. Postma, J. Roelofs, J. Goedhart, H. M. Looovers, A. J. Visser, P. J. Van Haastert, Sensitization of *Dictyostelium* chemotaxis by phosphoinositide-3-kinase-mediated self-organizing signalling patches. *J. Cell Sci.* **117**, 2925–2935 (2004).

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